Question ID: d1b66ae6

Question

$$-x+y=-3.5$$
$$x+3y=9.5$$

If (x, y) satisfies the system of equations above, what is the value of y ?

Question ID: ff501705

Question

$$\frac{3}{2}y - \frac{1}{4}x = \frac{2}{3} - \frac{3}{2}y$$

$$\frac{1}{2}x + \frac{3}{2} = py + \frac{9}{2}$$

In the given system of equations, p is a constant. If the system has no solution, what is the value of p?

Question ID: f75bd744

Question

$$4x - 6y = 10y + 2$$

$$ty = \frac{1}{2} + 2x$$

In the given system of equations, t is a constant. If the system has no solution, what is the value of t?

Question ID: 70feb725

Question

During a month, Morgan ran *r* miles at 5 miles per hour and biked *b* miles at 10 miles per hour. She ran and biked a total of 200 miles that month, and she biked for twice as many hours as she ran. What is the total number of miles that Morgan biked during the month?

Answer

A. 80

B. 100

C. 120

D. 160

Question ID: e1248a5c

Question

In the system of equations below, a and c are constants.

$$\frac{1}{2}x+\frac{1}{3}y=\frac{1}{6}$$
$$ax+y=c$$

If the system of equations has an infinite number of solutions (x,y), what is the value of a ?

Answer

A. $-\frac{1}{2}$

B. 0

C. $\frac{1}{2}$

D. $\frac{3}{2}$

Question ID: 52cb8ea4

Question

$7x - 5y = 4$

$4x - 8y = 9$

If (x, y) is the solution to the system of equations above, what is the value of $3x + 3y$?

Answer

A. -13

B. -5

C. 5

D. 13

Question ID: d7bf55e1

Question

A movie theater sells two types of tickets, adult tickets for $12 and child tickets for $8. If the theater sold 30 tickets for a total of $300, how much, in dollars, was spent on adult tickets? (Disregard the $ sign when gridding your answer.)

Question ID: f718c9cf

Question

$$5x + 14y = 45$$

$$10x + 7y = 27$$

The solution to the given system of equations is (x, y). What is the value of xy?

Question ID: 466b87e3

Question

$$y = \frac{1}{2}x + 8$$

$$y = cx + 10$$

In the system of equations above, c is a constant. If the system has no solution, what is the value of c ?

Question ID: e2e3942f

Question

$y = 2x + 1$

$y = ax - 8$

In the system of equations above, a is a constant. If the system of equations has no solution, what is the value of a ?

Answer

A. $-\frac{1}{2}$

B. 0

C. 1

D. 2

Question ID: 1e0a46e4

Question

Which system of linear equations has no solution?

Answer

A. $-2x + 3y = -9$

$2x - 3y = 9$

B. $2x - 3y = 9$

$3x + 4y = 10$

C. $2x - 3y = 9$

$-6x + 9y = -27$

D. $-2x + 3y = 9$

$4x - 6y = 18$

Question ID: 1e11190a

Question

Store A sells raspberries for $\$5.50$ per pint and blackberries for $\$3.00$ per pint. Store B sells raspberries for $\$6.50$ per pint and blackberries for $\$8.00$ per pint. A certain purchase of raspberries and blackberries would cost $\$37.00$ at Store A or $\$66.00$ at Store B. How many pints of blackberries are in this purchase?

Answer

A. 4

B. 5

C. 8

D. 12

Question ID: fb5e7f59

Question

$$-x - wy = -337$$

$$2x - wy = 47$$

In the given system of equations, w is a constant. In the *xy*-plane, the graphs of these equations intersect at the point $(q, 19)$, where q is a constant. What is the value of w?

Question ID: 567ac7ab

Question

One of the two equations in a linear system is $2x+6y=10$. The system has no solution. Which of the following could be the other equation in the system?

Answer

A. $x+3y=5$

B. $x+3y=-20$

C. $6x-2y=0$

D. $6x+2y=10$